QUICK CART-CASE STUDY

Orders & Support Tickets: Data Cleaning and Insights Report

Data quality issues found in orders.csv and support_tickets.csv, how each was fixed, and what the clean data reveals.

Working prototype : https://quickcart-casestudy.streamlit.app/

Part 1: Data Cleaning

Summary

orders.csv: 6,500 raw rows → 6,200 clean orders.

support_tickets.csv: 5,550 raw rows → 5,200 clean tickets.

All counts below refer to the final, de-duplicated records. The cleaning was done in code, so it can be re-run and every count independently re-checked.

1. orders.csv

Column	Issue Found	Example	Rows	Fix Applied
All columns	Test / synthetic records mixed in with real orders	ORD-999011, CUST-TEST11, amount −999.00	45	Removed
All columns	Exact duplicate rows	Identical copies of the same order	22	Removed; first copy kept
order_id	Same order repeated, with only the date written differently	ORD-700097 appears twice	233	De-duplicated on order_id (no other field differed)
order_date	Dates stored as text in 9 different formats	02/01/2026 · 07-Jan-2026 05:19 PM · 2026/02/12 19:20:43	6,200	Converted every format to one date/time; slash dates confirmed as DD/MM
order_amount	Amounts stored as text with currency symbols	₹1,999.00 · INR 699	1,113	Removed ₹ / INR / commas and converted to numbers
product_category	12 spellings for 4 categories (case, spaces, underscores)	ELECTRONICS · electronics · fashion_apparel	886	Mapped to 4 standard names
courier_partner	8 spellings for 4 couriers	' shadowfax_local ' · bluedart_air	775	Mapped to 4 standard names
customer_city	12 spellings for 6 cities, including synonyms	Bangalore / Bengaluru · New Delhi / Delhi NCR · 'mumbai '	1,034	Trimmed, standardised case, merged synonyms

> **Reconciliation:** 6,500 raw − 45 test records − 22 exact duplicates − 233 re-formatted duplicates = 6,200 clean orders.

No issues found in: customer_id, shipping_fee, payment_method and delivery_status (all values valid).

2. support_tickets.csv

Column	Issue Found	Example	Rows	Fix Applied
All columns	Exact duplicate rows	Identical copies of the same ticket	33	Removed; first copy kept
ticket_id	Same ticket repeated, with only the timestamps written differently	TKT-900016 appears twice	317	De-duplicated on ticket_id (no other field differed)
order_id	Malformed order references, so tickets could not be matched to orders	' ord_701234 ' · 701234 (no prefix)	624	Standardised to ORD-xxxxxx; every ticket now matches an order
ticket_created_at	Timestamps stored as text in 9 different formats	Feb 22, 2026 20:24 · 09.02.2026 21:20	5,200	Converted every format to one date/time
ticket_resolved_at	Timestamps stored as text in 9 different formats	01-Mar-26 12:22 · 2026/02/25 05:33:00	4,618	Converted every format; the 582 blanks are open tickets (left blank)
ticket_resolved_at	Resolved time earlier than created time (negative duration)	−2.9 to −17.6 hours	38	Flagged and excluded from resolution-time averages
refund_amount	Refunds stored as text with ₹ / Rs. / commas	₹3,499.00 · Rs. 1499	1,120	Removed symbols and converted to numbers
refund_amount	Missing refund written as blank or NIL	NIL · (blank)	562	Set to ₹0 (only occurs on no-refund, replacement or open tickets)
csat_score	Scores outside the 1–5 scale	99 · −1	30	Set to missing (invalid response)
csat_score	Blank: customer did not rate	(blank)	2,213	Kept as missing; excluded from the average, not counted as 0

> **Reconciliation:** 5,550 raw − 33 exact duplicates − 317 re-formatted duplicates = 5,200 clean tickets. Valid CSAT ratings: 5,200 − 30 invalid − 2,213 blank = 2,957.

No issues found in: customer_id (matches the order's customer on every ticket), issue_category, channel and resolution_status.

3. Why These Fixes Matter

- **Duplicates and test records** would have inflated order counts, order value and refund totals.
- **The 624 broken order references** would have dropped 12% of tickets from the courier and refund-policy analysis.
- **Invalid CSAT scores (99)** would have pushed the average above the 5-point maximum.
- **Negative resolution times** would have understated how long customers wait.
- **Inconsistent spellings** would have split one courier, city or category into several, distorting every breakdown.

Part 2: Summary and Insights

Key Findings at a Glance

- **Scale: 6,200** unique orders worth **₹2.29 Cr** (GMV, excluding shipping) and **5,200** unique support tickets, of which **3,404** ended in a refund or compensation.
- **Courier bottleneck:** Shadowfax loses or delays about **31%** of its orders (others: **18–19%**) and takes **32.6 h** to resolve tickets, around **70%** longer than the other couriers.
- **Support SLA:** average resolution time is **24.6 h** (median **20.0 h**); **40.3%** of tickets are resolved after **24 h**, **582** are still open, and average CSAT is **3.70** out of 5.
- **Refund leakage: 149** tickets were over-refunded, costing **₹4.68 L** (**4.7%** of all refunds); unpaid COD refunds are the largest source.
- **Further risks: 928** full refunds on delivered orders (**₹34.66 L**) and very low CSAT (**1.97**) on tickets that take more than **48 h**.

A. Overall Numbers

Objective: *After cleaning, how many unique orders and unique tickets are there? What are the true Total Order Value (GMV in ₹) and Total Refunds Issued?*

After cleaning, the dataset contains **6,200** unique orders and **5,200** unique support tickets. Total order value (GMV) is **₹2.29 Cr**, excluding shipping, and **3,404** tickets were refunded or compensated.

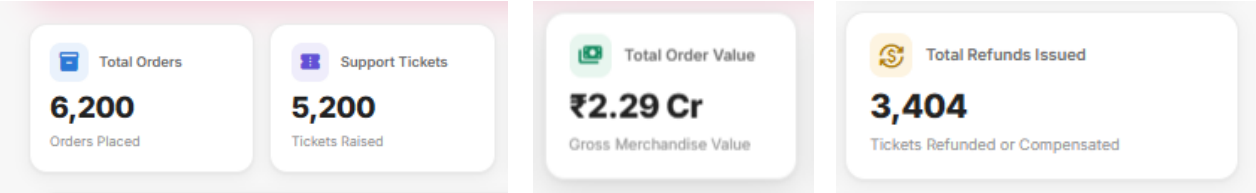


Figure 1: Headline numbers after cleaning

B. Courier Bottleneck

Objective: *Which courier_partner has the highest Lost/Delayed delivery rate and the slowest support resolution time?*

Shadowfax_Local is the clear outlier. In the scatter plot below it sits far from the other three couriers, with both the highest lost-and-delayed rate and the longest average resolution time. The other three couriers cluster together at about **18–19%** and **19 h**.

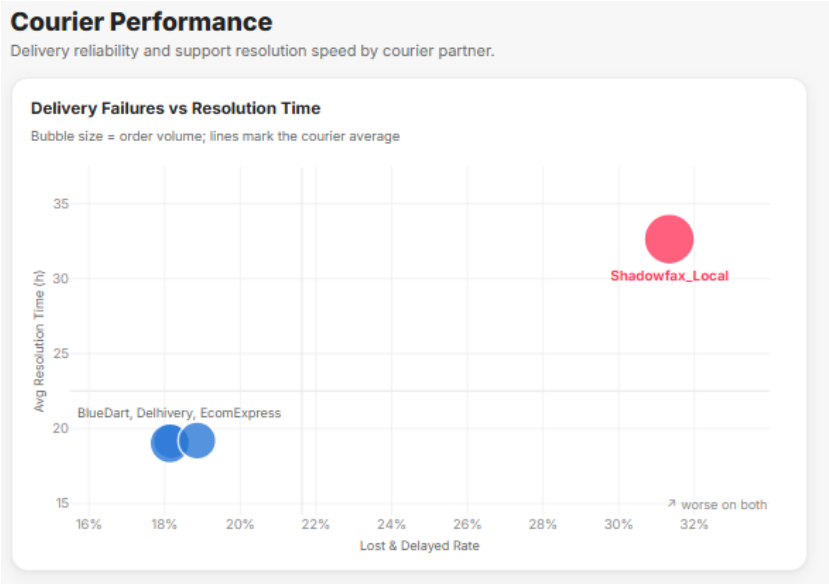
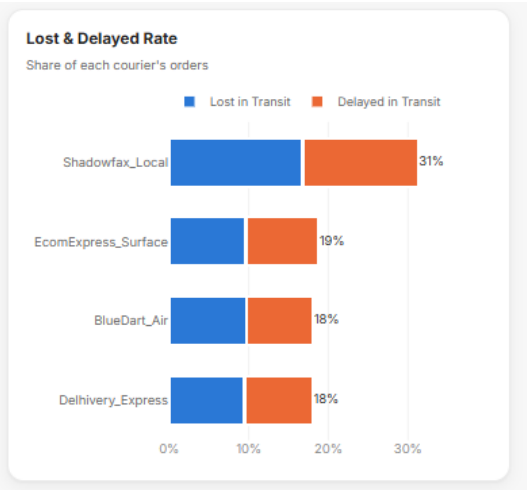
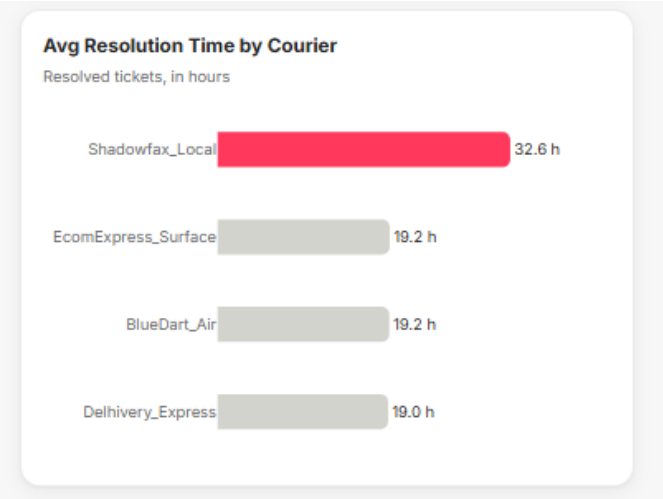


Figure 2: Delivery failures vs average resolution time (bubble size = order volume)

The courier-wise charts confirm this. About **31%** of Shadowfax orders are lost or delayed in transit, the highest share of any courier, split roughly evenly between lost and delayed. Its average resolution time is **32.6 h**, around **70%** higher than the other couriers (about **19 h**).





Figures 3 and 4: Lost & delayed rate (left) and average resolution time (right), by courier

> **Takeaway:** Shadowfax is the company's courier bottleneck: it is the worst on delivery reliability and on support resolution time.

C. Support SLA and CSAT

Objective: *What is the true average resolution time (in hours) for resolved tickets, how many tickets are still open, and what is the true average CSAT score (1–5)?*

The average resolution time for resolved tickets is **24.6 h**, with a median of **20.0 h**, and **40.3%** of resolved tickets take longer than **24 h**. In total, **582** tickets are still open. Valid CSAT ratings exist for **2,957** tickets (**57%** of all tickets), and their average is **3.70** out of 5.

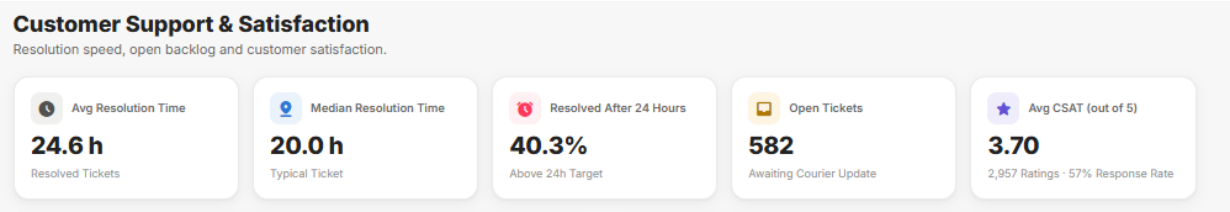


Figure 5: Support and satisfaction metrics

Satisfaction and speed differ sharply by channel. In-app chat produces the happiest customers and the fastest resolutions (**15.2 h**). Email and phone have the lowest satisfaction and the slowest resolutions (**38.6 h** and **37.8 h** respectively), and web helpdesk sits in between at **30.8 h**.

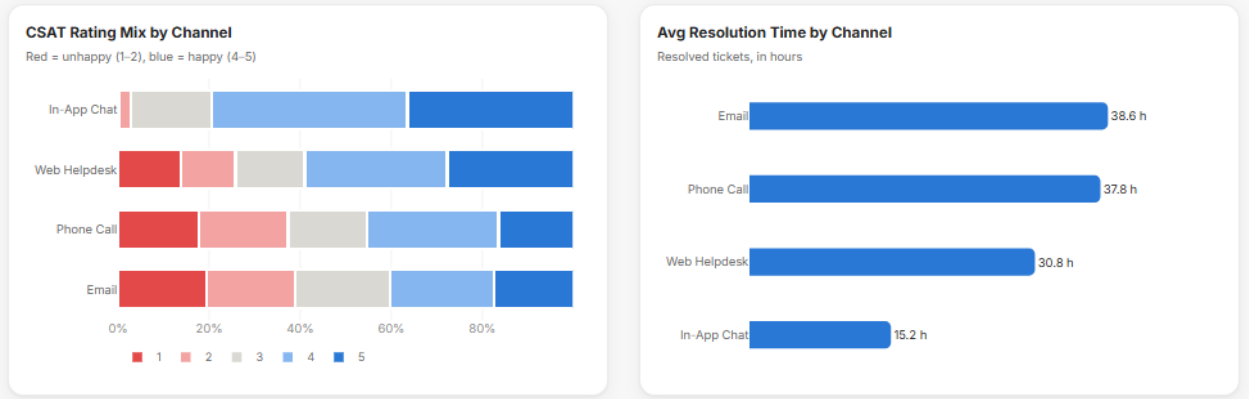


Figure 6: CSAT rating mix (left) and average resolution time (right), by support channel

> **Takeaway:** Email and phone are slow and leave customers less satisfied. Moving more contacts to in-app chat, or bringing email and phone resolution closer to chat speed, would improve both SLA and CSAT.

D. Refund Leak Check

Objective: *Compare the tickets against sop_policy_rules. How many tickets, and how much money (₹), were lost to refunds that broke the policy?*

A total of **149** tickets were refunded more than the policy allows. The excess comes to **₹4,68,054** (about **₹4.68 L**), which is **4.7%** of all refunds issued and could have been avoided.



Figure 7: Refund leakage summary

Leakage was measured against five policy rules. A ticket that breaks several rules is counted once, at its largest over-payment.

Policy Rule	What Counts as Leakage	Amount Counted
R1a Late support bonus	₹100 voucher added although the ticket was resolved within 24 hours	The extra ₹100
R1b Delayed orders still on the way	Refund above the ₹100 apology voucher	Amount above ₹100
R2 Damaged item: replacement or refund	Refund paid on top of a replacement	Full refund
R3 Unpaid COD order cancelled before shipping	Any refund paid	Full refund (customer never paid)
R4 Refund limit	Refund above order value + ₹100	Amount above the limit

The chart below shows how each rule adds to the total. Refunds on unpaid COD orders cancelled before shipping (R3) contribute the most, at **₹2.49 L** across 55 tickets. Replacement-plus-refund cases (R2) add **₹1.33 L** across 40 tickets, and refunds above the limit (R4) add **₹86,149** across 50 tickets. The late bonus rule (R1a) is minor at **₹400** across 4 tickets.

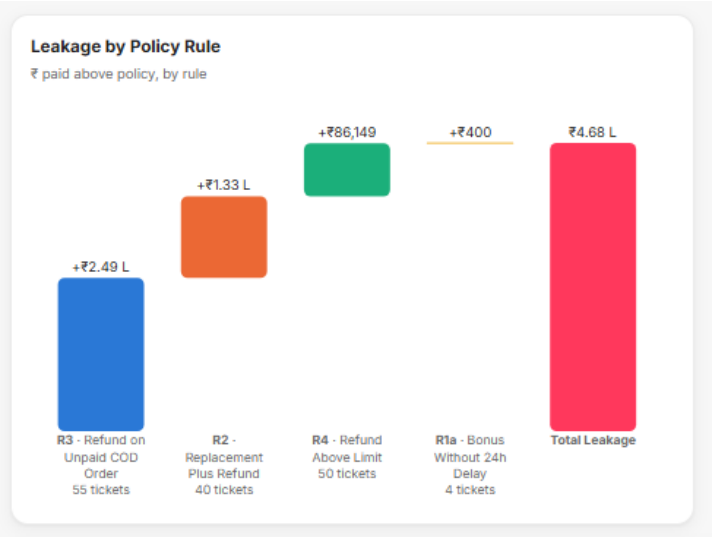


Figure 8: Leakage by policy rule

Additional Insights

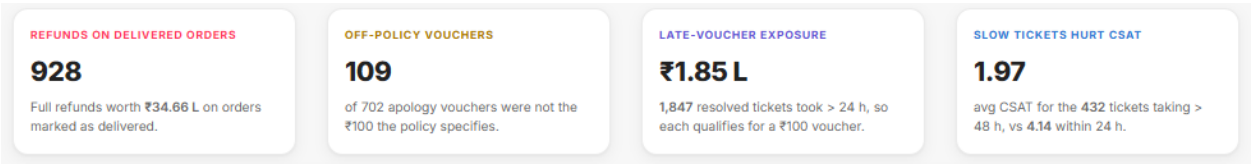


Figure 9: Additional risk indicators

1. **Refunds on delivered orders: 928** full refunds, worth **₹34.66 L**, were issued on orders marked as delivered. This can hurt revenue, so delivery damage claims and double payments through the payment gateway should be checked.
2. **Late-voucher exposure: 1,847** resolved tickets took more than **24 h**, and each qualifies for a **₹100** voucher, an exposure of **₹1.85 L**. Reducing turnaround time (TAT) would avoid most of this cost. In addition, **109** of **702** apology vouchers were not the **₹100** the policy specifies.

3. **Slow tickets hurt CSAT:** the **432** tickets that took more than **48 h** average a CSAT of only **1.97**, compared with **4.14** for tickets resolved within **24 h**.
4. **Shadowfax drives leakage too:** because most lost and delayed orders come from Shadowfax, it also contributes the most to refund leakage (**₹1.79 L**, roughly **38%** of the total), ahead of Delhivery (**₹1.20 L**), EcomExpress (**₹1.12 L**) and BlueDart (**₹56,860**).

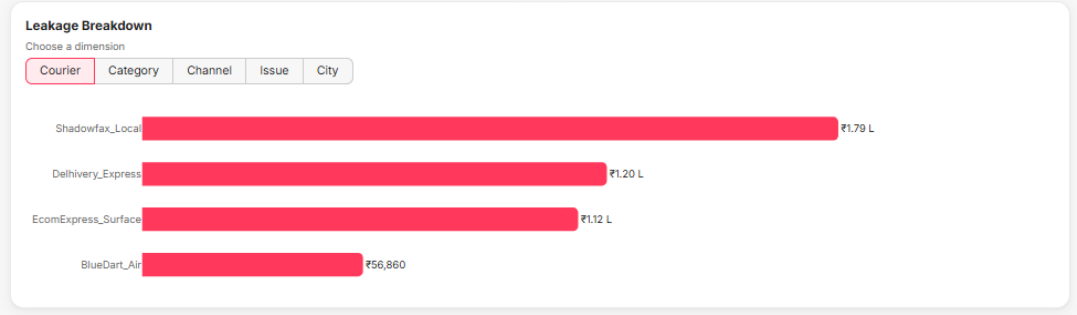


Figure 10: Refund leakage by courier

Recommendations and Actions

1. Shadowfax_Local is failing

- **Issue:** Carries **37%** of orders but loses or delays **31.3%** of them (others ~**18%**), and its tickets take **32.6 h** to resolve vs ~**19 h**.
- **Fix:** Move high-value orders to other couriers and add an SLA with penalties and claims for lost parcels.

2. Refunds break policy

- **Issue: 149** tickets were over-refunded by **₹4.68 L**, including refunds on unpaid COD orders and replacement plus refund.
- **Fix:** Build the 4 refund rules into the refund tool as hard blocks, with manager approval for exceptions.

3. Unverified refunds on delivered orders

- **Issue: ₹34.66 L** was refunded on **928** orders marked as delivered (damaged and double-charged claims).
- **Fix:** Check double charges with the payment gateway and require photo proof for damage, offering a replacement first.

4. Slow support

- **Issue: 40%** of tickets take over **24 h**, and CSAT falls from 4.1 to 2.0 after **48 h**. Email and phone are twice as slow as chat.
- **Fix:** Steer customers to chat, let the chatbot resolve lost and delayed cases instantly, and auto-escalate at **18 h**.

5. Too many tickets

- **Issue: 84%** of orders raise a ticket, mostly “where is my order”, and **582** are stuck awaiting courier updates.
- **Fix:** Connect courier tracking and notify customers of delays before they ask, with automatic refunds for lost parcels.